

University of Cape Town

Bayesian Computational Methods

Assignment

Help

Author:
Isabella Lethbridge
Tayyeb Abrahams

Student Number:
LTHISA001
ABRMOE030

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Introduction

Question a

Each of the n birds is caught exactly once. The capture day of bird i is denoted by t_i and is treated as fixed. Every bird moults at the same constant rate, with moult lasting τ days for all birds. The start of moult for bird i, T_i, is an unobserved random variable, where

$$T_i \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2), \qquad f_T(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right), \qquad i = 1, \dots, n. \tag{1}$$

Let $Z = \frac{T_i - \mu}{\sigma} \sim \mathrm{N}(0,1)$ and let Φ denote the corresponding CDF. In the Type 1 model, only the status ($s_i \in \{1,2,3\}$) of each bird is observed.

For $s_i = 3$, bird i has not yet started moult:

$$P(t_i) = \Pr(T_i > t_i) = \Pr\left(Z > \frac{t_i - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{t_i - \mu}{\sigma}\right). \tag{2}$$

For $s_i = 2$, bird i is currently moulting:

$$Q(t_i) = \Pr(t_i - \tau \le T_i \le t_i) = \Pr\left(\frac{t_i - \tau - \mu}{\sigma} \le Z \le \frac{t_i - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \mu}{\sigma}\right) - \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right). \tag{3}$$

For $s_i = 1$, bird i has finished moulting:

$$R(t_i) = \Pr(T_i \le t_i - \tau) = \Pr\left(Z \le \frac{t_i - \tau - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right). \tag{4}$$

Given t_i, the status s_i is a single categorical draw with probabilities $P(t_i), Q(t_i)$ and $R(t_i)$, given in Equation 2, Equation 3 and Equation 4. Thus, the conditional probability of s_i can be written as

$$\Pr(s_i \mid t_i, \mu, \sigma^2, \tau) = P(t_i)^{\mathrm{I}\{s_i=3\}}\, Q(t_i)^{\mathrm{I}\{s_i=2\}}\, R(t_i)^{\mathrm{I}\{s_i=1\}}. \tag{5}$$

Since the birds are independent, the likelihood is the product of Equation 5 over all n birds

$$\begin{aligned} L(\boldsymbol{s}, \boldsymbol{t}, \mu, \sigma^2, \tau) &= \prod_{i=1}^{n} P(t_i)^{\mathrm{I}\{s_i=3\}}\, Q(t_i)^{\mathrm{I}\{s_i=2\}}\, R(t_i)^{\mathrm{I}\{s_i=1\}} \\ &= \prod_{i=1}^{n} \left[1 - \Phi(\frac{t_i - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=3\}} \left[\Phi(\frac{t_i - \mu}{\sigma}) - \Phi(\frac{t_i - \tau - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=2\}} \left[\Phi(\frac{t_i - \tau - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=1\}}. \end{aligned} \tag{6}$$

On any capture day t, the population consists of birds which belong to three different states.

Birds have either not yet started to moult, are undergoing moult or have completed moult. These states occur in proportions $P(t), Q(t)$ and $R(t)$, which depend on t through μ, σ^2 and τ. Thus, the moult index has a mixture distribution with a point mass at 0 and 1 with weights $P(t)$ and $R(t)$, respectively, as well as a continuous component on $(0, 1)$ with weight $Q(t)$. In the Type 1 model, the moult index is not observed. We only observe the state that bird i belongs to. Thus, each bird contributes the weight of the state it belongs to $P(t_i)$, $Q(t_i)$ or $R(t_i)$, to Equation 6.

Question b

In the Type 2 model, the moult scores (y_i) for birds that are currently moulting are observed. The birds are grouped according to their states. Let $\mathcal{F} = \{i : s_i = 1\}$ denote the set of birds that have already moulted, $\mathcal{M} = \{i : s_i = 2\}$ denote the set of birds that are undergoing moult and $\mathcal{N} = \{i : s_i = 3\}$ denote the set of birds that have not yet started to moult. Let $m = |\mathcal{M}|$ be the number of birds currently moulting.

For a bird undergoing moult, the linear assumption gives $T_i = t_i - y_i\tau$. The probability density function of its observed moult score is

$$f_y(y_i) = \tau f_T(t_i - y_i\tau) \qquad 0 < y_i < 1. \tag{7}$$

The likelihood contributions of birds that have already moulted and birds that have not yet started to moult are the same as in the Type 1 model. Thus, the Type 2 likelihood is

$$L(\boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \mu, \sigma^2, \tau) = \prod_{i\in\mathcal{F}} R(t_i) \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{N}} P(t_i), \tag{8}$$

and the joint posterior distribution is

$$\pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto \prod_{i\in\mathcal{F}} R(t_i) \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{N}} P(t_i)\, \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau), \tag{9}$$

with the following prior distributions

$$\mu \sim \text{Uniform}(100, 160), \quad \sigma^2 \sim \text{IG}(1, 1), \quad \tau \sim \text{Uniform}(50, 180). \tag{10}$$

Data augmentation

The probabilities $R(t_i)$ and $P(t_i)$ in Equation 9 are integrals over the unobserved start date

$$\begin{aligned} R(t_i) &= \Pr(T_i \le t_i - \tau) = \int_{-\infty}^{\infty} f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\}\, dT_i, \\ P(t_i) &= \Pr(T_i > t_i) = \int_{-\infty}^{\infty} f_T(T_i)\, \mathrm{I}\{T_i > t_i\}\, dT_i. \end{aligned} \tag{11}$$

The status is fully determined by the start date (T_i), so each integrand in Equation 11 is the joint density of the start date and the bird's status

$$\begin{aligned} p(T_i, s_i = 1 \mid \mu, \sigma^2, \tau) &= f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\}, \\ p(T_i, s_i = 3 \mid \mu, \sigma^2) &= f_T(T_i)\, \mathrm{I}\{T_i > t_i\}. \end{aligned} \tag{12}$$

T_i for $i \in \mathcal{F} \cup \mathcal{N}$ is treated as a latent variable and samples are drawn from the augmented posterior

$$\begin{aligned} \pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto & \prod_{i\in\mathcal{F}} \left[f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\} \right] \prod_{i\in\mathcal{M}} \left[\tau f_T(t_i - y_i\tau) \right] \\ & \times \prod_{i\in\mathcal{N}} \left[f_T(T_i)\, \mathrm{I}\{T_i > t_i\} \right] \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau). \end{aligned} \tag{13}$$

Integrating out each T_i in Equation 13 returns $R(t_i)$ or $P(t_i)$, so that

$$\int \pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t})\, d\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})} = \pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}). \tag{14}$$

The posterior distribution of interest, Equation 9, is the marginal distribution of the augmented posterior. Thus, the draws of (μ, σ^2, τ) from a sampler targeting the augmented posterior are draws from the required posterior. No latent variables are needed for the moulting birds, since $T_i = t_i - y_i\tau$ is a deterministic function of τ for $i \in \mathcal{M}$. Since $0 < y_i < 1$, the condition $t_i - \tau \le T_i \le t_i$ is always satisfied.

Conditional distributions of T_i

The conditional distributions of the latent start dates are the same truncated normal distributions as in the Type 1 Gibbs sampler:

$$\begin{aligned} T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i \le t_i - \tau\}, \quad i \in \mathcal{F}, \\ T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i > t_i\}, \qquad i \in \mathcal{N}. \end{aligned} \tag{15}$$

Conditional distribution of μ

From Equation 13, the full conditional distribution of μ is

$$
\begin{aligned}
\pi(\mu \mid \cdot) &\propto \prod_{i \in \mathcal{F}} [f_T(T_i)] \prod_{i \in \mathcal{M}} [f_T(t_i - y_i\tau)] \prod_{i \in \mathcal{N}} [f_T(T_i)]\ \pi(\mu) \\
&\propto \prod_{i \in \mathcal{F}} [f_T(T_i)] \prod_{i \in \mathcal{M}} [f_T(T_i)] \prod_{i \in \mathcal{N}} [f_T(T_i)]\ \pi(\mu) \\
&\propto \prod_{i=1}^{n} f_T(T_i)\, \pi(\mu) \\
&= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(T_i - \mu)^2}{2\sigma^2}\right) \mathrm{I}\{100 \le \mu \le 160\} \\
&\propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (T_i - \mu)^2\right) \mathrm{I}\{100 \le \mu \le 160\}.
\end{aligned}
$$

Let $\bar{T} = \frac{1}{n}\sum_{i=1}^{n} T_i$. Expanding the sum of squares gives

$$
\begin{aligned}
\sum_{i=1}^{n} (T_i - \mu)^2 &= \sum_{i=1}^{n} \left[(T_i - \bar{T}) + (\bar{T} - \mu)\right]^2 \\
&= \sum_{i=1}^{n} (T_i - \bar{T})^2 + 2(\bar{T} - \mu) \sum_{i=1}^{n} (T_i - \bar{T}) + n(\bar{T} - \mu)^2 \\
&= \sum_{i=1}^{n} (T_i - \bar{T})^2 + n(\bar{T} - \mu)^2,
\end{aligned}
$$

since $\sum_{i=1}^{n} (T_i - \bar{T}) = 0$. Thus,

$$
\pi(\mu \mid \cdot) \propto \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{T})^2\right) \mathrm{I}\{100 \le \mu \le 160\}.
$$

This is the kernel of a truncated normal distribution

$$
\mu \mid \cdot \sim \mathrm{N}\left(\bar{T}, \frac{\sigma^2}{n}\right) \mathrm{I}\{100 \le \mu \le 160\}. \tag{16}
$$

For the moulting birds, $T_i = t_i - y_i\tau$ is used in $\bar{T}$.

Conditional distribution of σ^2

Let $S = \sum_{i=1}^{n} (T_i - \mu)^2$. From Equation 13, the full conditional distribution of σ^2 is

$$\begin{aligned}
\pi(\sigma^2 \mid \cdot) &\propto \prod_{i\in\mathcal{F}} [f_T(T_i)] \prod_{i\in\mathcal{M}} [f_T(t_i - y_i\tau)] \prod_{i\in\mathcal{N}} [f_T(T_i)]\ \pi(\sigma^2) \\
&\propto \prod_{i=1}^{n} f_T(T_i)\, \pi(\sigma^2) \\
&\propto (\sigma^2)^{-n/2} \exp\left(-\frac{S}{2\sigma^2}\right) (\sigma^2)^{-2} \exp\left(-\frac{1}{\sigma^2}\right) \\
&= (\sigma^2)^{-(\frac{n}{2}+1)-1} \exp\left(-\frac{1+\frac{S}{2}}{\sigma^2}\right).
\end{aligned}$$

This is the kernel of an inverse-gamma distribution

$$\sigma^2 \mid \cdot \sim \text{IG}\left(\frac{n}{2}+1,\ 1+\frac{1}{2}\sum_{i=1}^{n}(T_i-\mu)^2\right). \tag{17}$$

Equation 17 differs from the Type 1 conditional given in the assignment, which corresponds to the prior $\pi(\sigma^2) \propto 1/\sigma^2$ as opposed to $\text{IG}(1,1)$.

Conditional distribution of τ

From Equation 13, with $T_i = t_i - y_i\tau$ for the moulting birds, the full conditional distribution of τ is

$$\begin{aligned}
\pi(\tau \mid \cdot) &\propto \prod_{i\in\mathcal{F}} [\text{I}\{T_i \le t_i - \tau\}] \prod_{i\in\mathcal{M}} [\tau f_T(t_i - y_i\tau)]\ \pi(\tau) \\
&\propto \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{F}} \text{I}\{T_i \le t_i - \tau\}\, \text{I}\{50 \le \tau \le 180\}.
\end{aligned}$$

The moulting birds contribute

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\frac{1}{2\sigma^2}\sum_{i\in\mathcal{M}}(t_i - \mu - y_i\tau)^2\right).$$

Expanding the square gives

$$\sum_{i\in\mathcal{M}}(t_i - \mu - y_i\tau)^2 = \sum_{i\in\mathcal{M}}(t_i-\mu)^2 - 2\tau\sum_{i\in\mathcal{M}} y_i(t_i-\mu) + \tau^2\sum_{i\in\mathcal{M}} y_i^2,$$

where the first term does not depend on τ. Thus,

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right),$$

where $\beta = \frac{\sum_{i\in\mathcal{M}} y_i^2}{2\sigma^2}$ and $\gamma = \frac{\sum_{i\in\mathcal{M}} y_i(t_i-\mu)}{\sigma^2}$.

The finished birds only contribute through their indicators. The product equals one only if the condition holds for every finished bird, so

$$\prod_{i\in\mathcal{F}} \mathrm{I}\{T_i \le t_i - \tau\} = \mathrm{I}\{\tau \le \min_{i\in\mathcal{F}}(t_i - T_i)\}$$

Let $h = \min_{i\in\mathcal{F}}(t_i - T_i)$. Thus, the duration of moult cannot exceed the time between the start date and the capture date of any finished bird.

Combining these two terms with the prior gives

$$\pi(\tau \mid \cdot) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right) \mathrm{I}\{50 \le \tau \le \min(180, h)\}.$$

This is the kernel of a Modified-Half-Normal distribution (Sun et al., 2021)

$$\tau \mid \cdot \sim \mathrm{MHN}\left(m+1,\ \frac{\sum_{i\in\mathcal{M}} y_i^2}{2\sigma^2},\ \frac{\sum_{i\in\mathcal{M}} y_i(t_i-\mu)}{\sigma^2}\right) \mathrm{I}\{50 \le \tau \le \min(180, h)\}. \tag{18}$$

In the Type 1 model, τ appears only in the indicators and has a uniform conditional distribution. In the Type 2 model, the observed moult scores result in a non-uniformly shaped conditional of τ. Sampling from Equation 18 can be simplified using a further data augmentation step (Lemma 5 of Sun et al., 2021). The form of this data augmentation depends on the sign of γ. Since $y_i > 0$, $\gamma \ge 0$ if and only if $\mu \le \bar{t}_y = \sum_{i\in\mathcal{M}} y_i t_i \big/ \sum_{i\in\mathcal{M}} y_i$.

When $\gamma \ge 0$, a latent variable V is introduced such that $V \mid \tau \sim \mathrm{Poisson}(\gamma\tau)$. Multiplying the conditional distribution of τ by the Poisson probability gives

$$\tau^m e^{-\beta\tau^2+\gamma\tau} \cdot \frac{e^{-\gamma\tau}(\gamma\tau)^V}{V!} \propto \tau^{m+V} e^{-\beta\tau^2},$$

so the term $e^{\gamma\tau}$ cancels. Substituting $W = \tau^2$ and noting that the indicator only involves τ,

$$W \mid V \sim \mathrm{Gamma}\left(\frac{m+1+V}{2},\ \beta\right) \mathrm{I}\{50^2 \le W \le \min(180, h)^2\}, \tag{19}$$

where β is the rate parameter. When $\gamma = 0$, $V = 0$ and Equation 19 reduces to a truncated $\mathrm{Gamma}\left(\frac{m+1}{2}, \beta\right)$ distribution.

When $\gamma < 0$, a latent variable U is introduced instead. Using the parameterisation $\mathrm{GIG}(p, a, b)$ with density proportional to $u^{p-1}\exp\{-(au + b/u)/2\}$,

$$\begin{aligned} U \mid \tau &\sim \mathrm{GIG}\left(\frac{1}{2},\ 1,\ \gamma^2\tau^2\right), \\ W \mid U &\sim \mathrm{Gamma}\left(\frac{m+1}{2},\ \beta + \frac{\gamma^2}{2U}\right) \mathrm{I}\{50^2 \le W \le \min(180, h)^2\}. \end{aligned} \tag{20}$$

Under this parameterisation of the GIG distribution the rate parameter is $\beta + \frac{\gamma^2}{2U}$. Sun et al. (2021) state the rate as $\beta + \frac{\gamma^2}{U}$, which corresponds to a different parameterisation of the GIG distribution.

In both cases, the truncated gamma distribution can be sampled exactly using the inverse CDF method, after which $\tau = \sqrt{W}$. The latent variable V or U is redrawn given the current τ at every iteration and then discarded, so the conditionals of the other parameters are unchanged.

The Gibbs sampling algorithm

1. Initialise $\mu^{(0)} \sim \mathrm{U}(100, 160)$, $\sigma^{2(0)} \sim \mathrm{IG}(1, 1)$ and $\tau^{(0)} \sim \mathrm{U}(50, 180)$. Draw $T_i^{(0)}$ for $i \in \mathcal{F} \cup \mathcal{N}$ from their truncated normal conditional distributions, Equation 15, given these values.
2. For $k = 1, \ldots, K$:
   a. For $i \in \mathcal{F}$, draw $T_i^{(k)} \sim \mathcal{N}(\mu^{(k-1)}, \sigma^{2(k-1)})\, \mathrm{I}\{T_i \leq t_i - \tau^{(k-1)}\}$.
   b. For $i \in \mathcal{N}$, draw $T_i^{(k)} \sim \mathcal{N}(\mu^{(k-1)}, \sigma^{2(k-1)})\, \mathrm{I}\{T_i > t_i\}$.
   c. For $i \in \mathcal{M}$, set $T_i^{(k)} = t_i - y_i \tau^{(k-1)}$.
   d. Draw $\mu^{(k)} \sim \mathrm{N}(\bar{T}^{(k)}, \sigma^{2(k-1)}/n)\, \mathrm{I}\{100 \leq \mu \leq 160\}$.
   e. Draw $\sigma^{2(k)} \sim \mathrm{IG}(\frac{n}{2} + 1,\ 1 + \frac{1}{2}\sum_{i=1}^{n} (T_i^{(k)} - \mu^{(k)})^2)$.
   f. Compute $h^{(k)} = \min_{i \in \mathcal{F}}(t_i - T_i^{(k)})$, and $\beta^{(k)}$ and $\gamma^{(k)}$ evaluated at $\mu^{(k)}$ and $\sigma^{2(k)}$. If $\gamma^{(k)} \geq 0$, draw $V \sim \mathrm{Poisson}(\gamma^{(k)}\tau^{(k-1)})$ and $W \sim \mathrm{Gamma}(\frac{m+1+V}{2}, \beta^{(k)})$; otherwise, draw $U \sim \mathrm{GIG}(\frac{1}{2}, 1, (\gamma^{(k)}\tau^{(k-1)})^2)$ and $W \sim \mathrm{Gamma}(\frac{m+1}{2}, \beta^{(k)} + \frac{(\gamma^{(k)})^2}{2U})$. In both cases, W is truncated to $[50^2, \min(180, h^{(k)})^2]$. Set $\tau^{(k)} = \sqrt{W}$.
3. After discarding a burn-in period, retain the draws $(\mu^{(k)}, \sigma^{2(k)}, \tau^{(k)})$ as samples from the posterior $\pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t})$. The latent variables $T_i^{(k)}$, V and U are discarded.

Question c

We sample from the likelihood surface using a Metropolis Hastings algorithm. However we have the problem that the steps of our random walk are drawn a multivariate normal distribution. But each of our three parameters do not have support over the entire real line: μ, which determines the average start time, is constrained by a uniform prior to be situated between 100–160; τ, which determines the speed of moulting, is also contrainted by its prior to be situated within 50–180; finally the variance is strictly positive. We define the following transformations:

$$
\begin{aligned}
f(\mu) &= \operatorname{logit} \frac{\mu - 100}{60} \\
g(\tau) &= \operatorname{logit} \frac{\tau - 50}{130} \\
h(\sigma^2) &= \log \sigma^2
\end{aligned}
$$

These transformation each have support over the entire real line; thus we may perform our random walk over this transformed space. Since we are unsure how our parameters are

correlated; to begin with, the steps of our random walk are drawn from:

$$e_i \sim \mathcal{N}(0, c\, I)$$

For some constant c. However, when back transform (let θ be our original parameters and ϕ their transformations) we must include the jacobian term in the posterior likelihood in order to ensure that the Metropolis-Hasting algorithm converges to the appropriate likelihood; the log-likelihood becomes:

$$L(\theta) + \log(\pi(\theta)) + \log|J(\phi)|$$

Where $J(\phi)$ is the Jacobian of our transformation. However, we would like to tune our random walk to have a more appropriate covariance matrix; to achieve this we utilize 3000 burn-in iterations and estimate the covariance matrix Σ of our transformed parameters on these iterations. Then each of our retained draws each have their random walk drawn from:

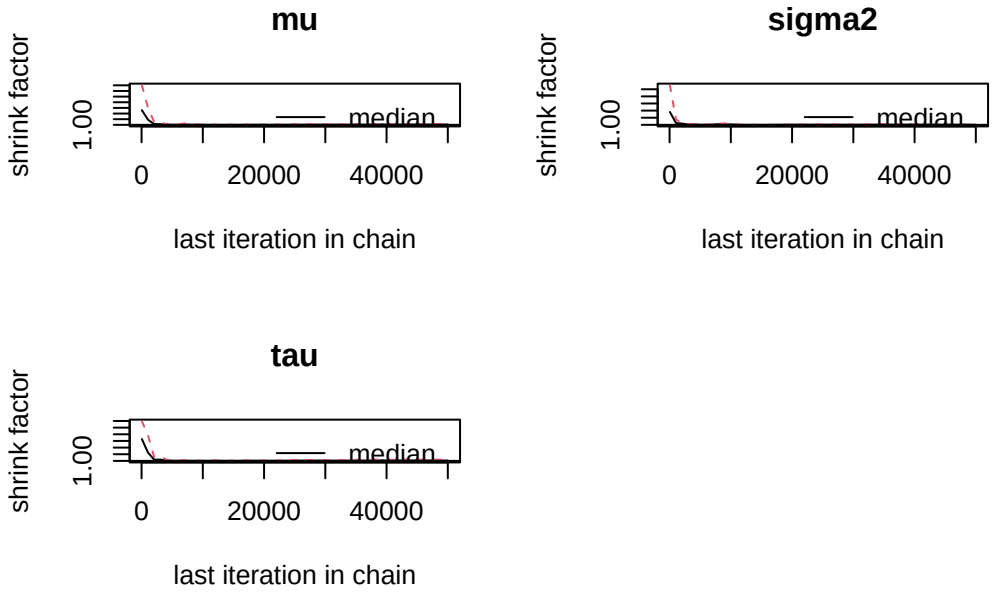
$$e \sim \mathcal{N}(0, \Sigma)$$

And the Metropolis Hastings algorithm algorithm proceeds as per usual.\

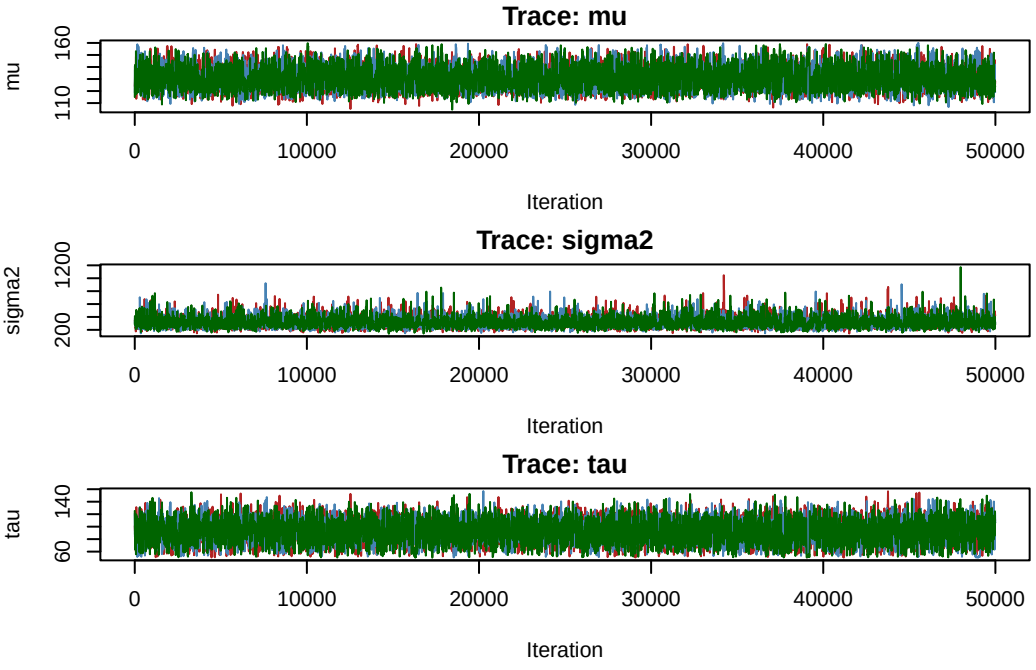
We will run 50,000 iterations on three chains which are each randomly initialized: μ and τ a drawn from uniform distributions and σ^2 drawn from IG$(1, 1)$. The initial values for our chains as well as the acceptance rates for our MH sampler can be seen below:

chain	burn_accept	main_accept	init_mu	init_sigma2	init_tau
1	0.21	0.27	141.92	1.65	87.14
2	0.21	0.34	115.56	0.76	114.27
3	0.20	0.23	109.01	2.55	74.15

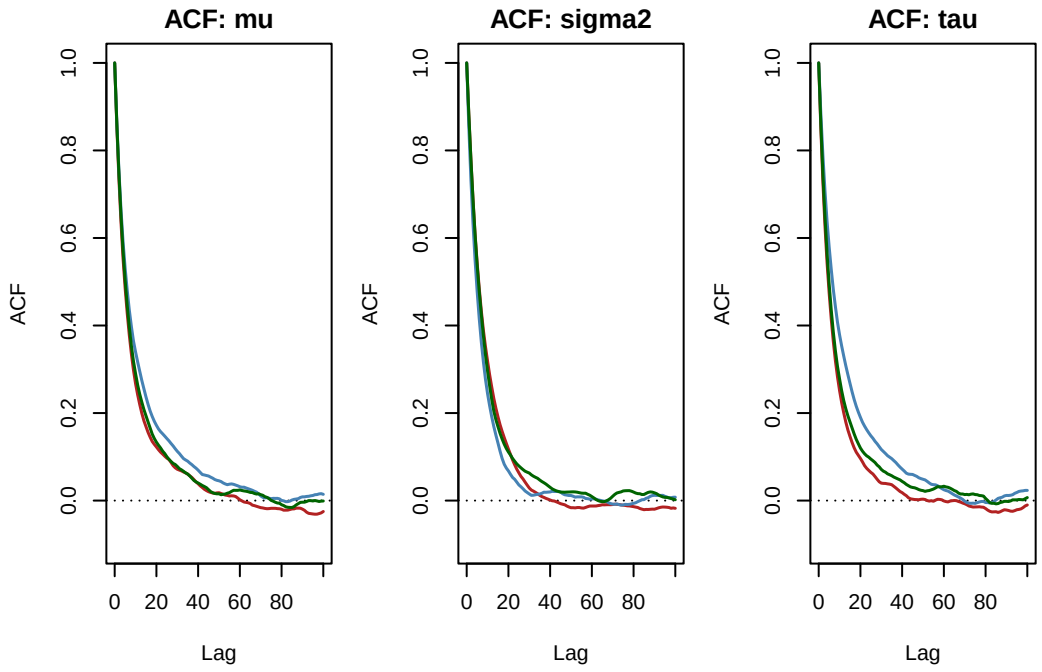
The acceptance rates of 0.21 suggests that the step sizes of our random walk are neither to small (which would results in high acceptance rates) nor too large (which would result in low acceptance rates by jumping into sparse areas of the density) and that our covariance matrix was adequately tuned in the burn-in draws. To assess the convergence of our chains obtain our $\hat{R}$ statistic and can inspect the gelman plot of our chains:



We find that $\hat{R} \approx 1$ for each of our parameters and conclude that our MH sampler has converged; alternatively we can inspect the trace plots of each of our parameters:



And conclude that the traceplots are stationary from visual inspection. We should note, however, that the effective sample size of our MH sampler is likely much smaller than the chosen number of iterations because of autocorrelation in the draws; inspection the ACF plots of our draws:



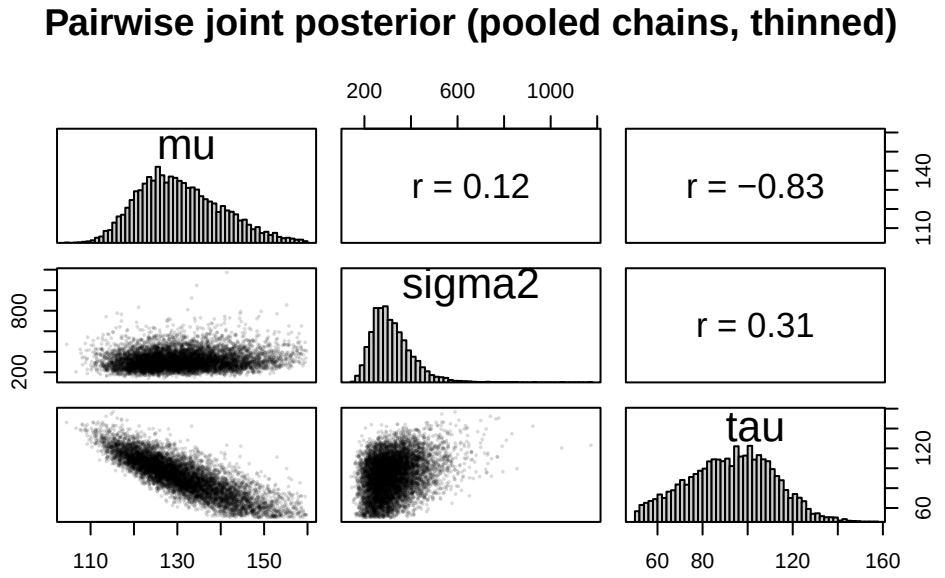
That is, it takes $\approx$ 100 iterations before draws become uncorrelated; for this reason, it's unsurprising that the effect sample sizes of our chains are:

```
      mu   sigma2      tau
8190.576 9340.338 8733.665
```

Which are much smaller than the number of retained draw. Finally, we obtain our posterior estimates and 95% credibility intervals for each of our three parameters:

Parameter	Mean	SD	Q2.5	Median	Q97.5	ESS
mu	130.70	9.54	114.82	129.63	151.54	8190.58
sigma2	319.19	88.67	193.13	303.90	532.89	9340.34
tau	92.68	19.13	55.85	93.53	128.55	8733.66

We can more directly inspect the marginal densities and bivariate distributions by the figures below:



We should note that the credibility intervals on our estimate for μ are quite wide and spans almost the entire interval defined by the prior, the posterior estimate of 130 is unsurprising given that almost all of the birds were caught on day 127; the outliers situated beyond the prior interval likely explains the wide credibility interval. The credibility interval for τ however appears to be notably narrower than its prior. We note that while μ and σ^2 appear to be broadly uncorrelated, τ is weakly correlated with σ^2 ($\rho = 0.31$) and strongly correlated with μ ($\rho = -0.81$); this is expected since τ and μ are related by $\mu = t - y\tau$, which also suggests that the spread in start dates scales with τ. # Question c

Table 3: Random starting values and acceptance rates of the three Metropolis-Hastings chains.

Chain	$\mu^{(0)}$	$\sigma^{2(0)}$	$\tau^{(0)}$	Acceptance rate
1	113.29	583.19	117.06	0.22
2	125.39	20.93	151.60	0.22
3	148.43	715.56	140.47	0.22

Table 4: Posterior summaries for the Type 3 model obtained using the tuned random walk Metropolis-Hastings algorithm (3 chains, 45 000 retained draws each).

	Mean	SD	2.5%	Median	97.5%	ESS	$\widehat{R}$
μ	130.53	9.36	114.83	129.50	150.77	7970	1
σ^2	316.78	84.13	194.36	303.44	513.66	8024	1
τ	92.76	18.94	55.42	93.72	127.81	8879	1
σ	17.65	2.27	13.94	17.42	22.66	8046	1

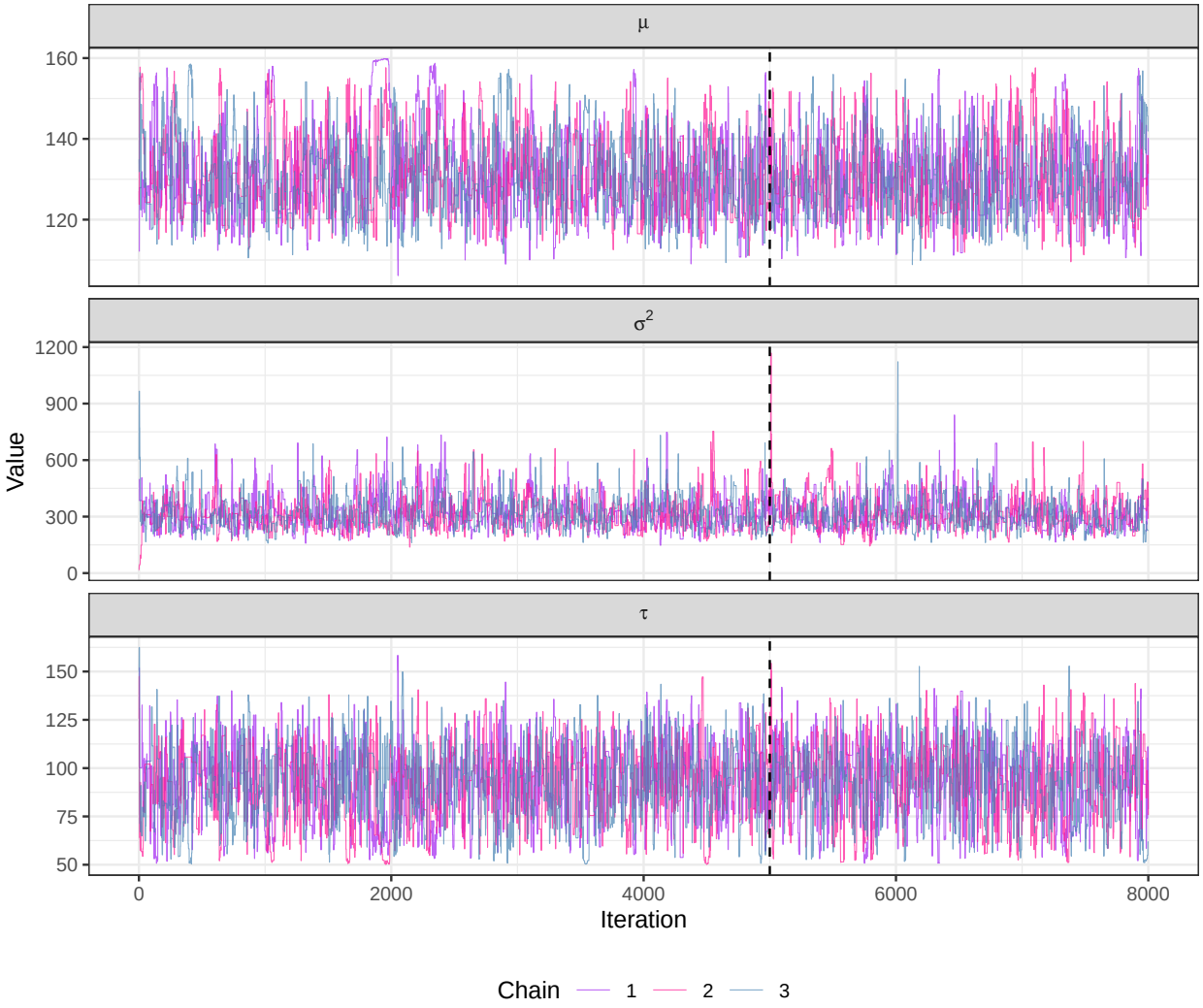


Figure 1: Trace plots of the three Metropolis-Hastings chains for the first 8 000 iterations. The burn-in is indicated by the dashed line.

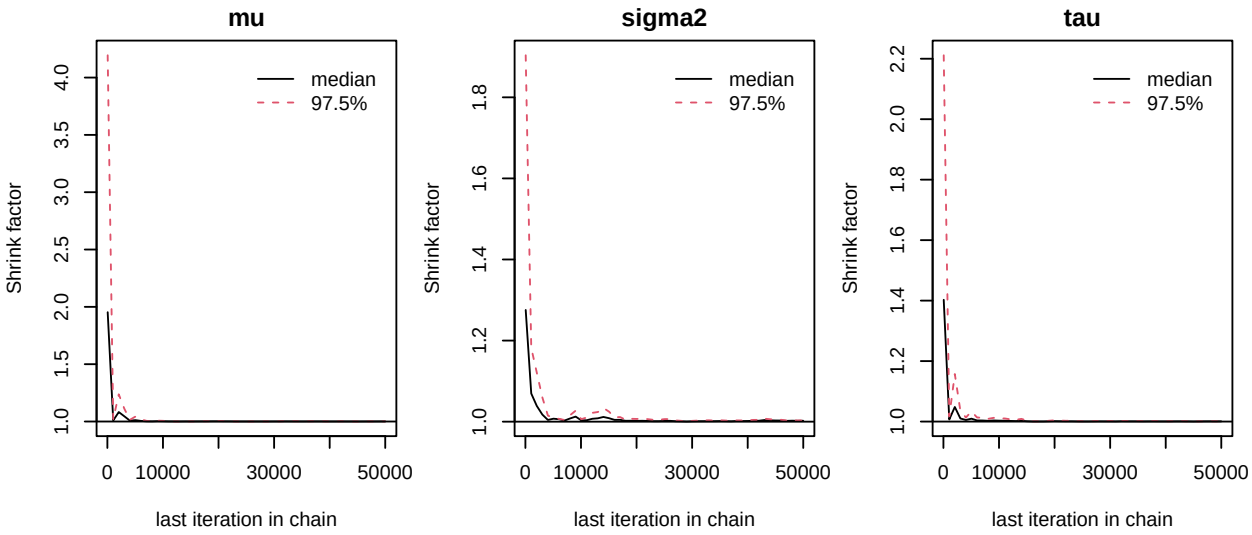


Figure 2: Gelman-Rubin shrink factor as a function of the number of iterations.

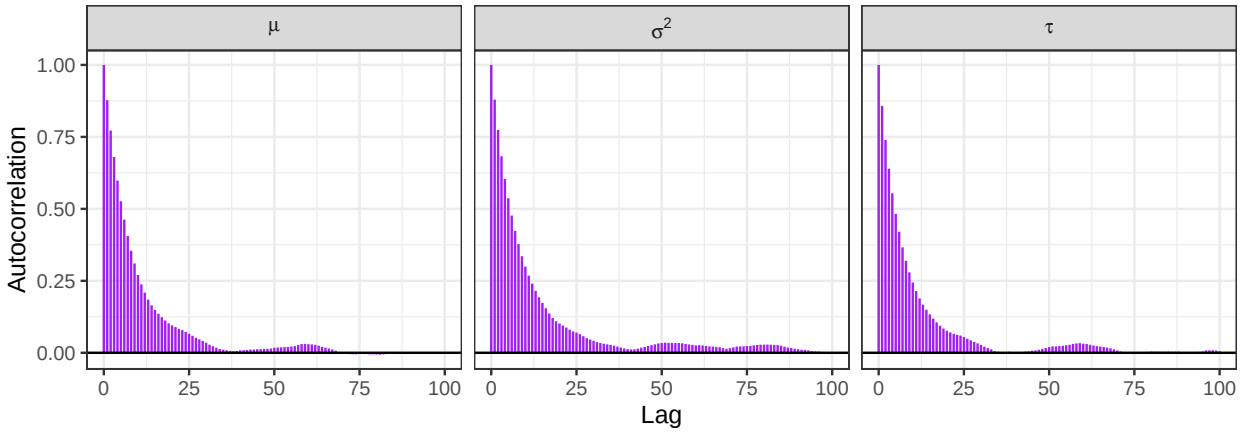


Figure 3: Autocorrelation of the retained draws of the first Metropolis-Hastings chain.

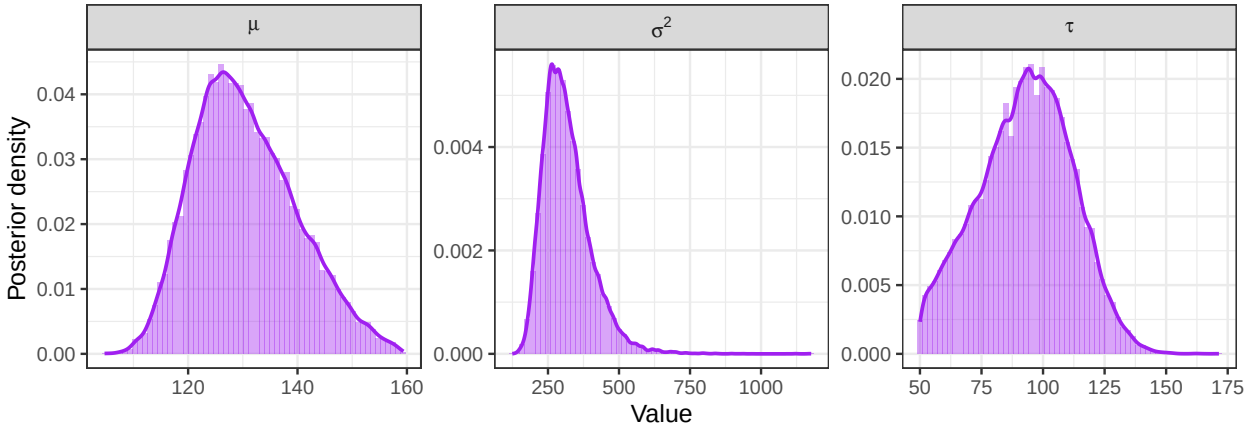


Figure 4: Marginal posterior distributions of the Type 3 model parameters from the Metropolis-Hastings draws.

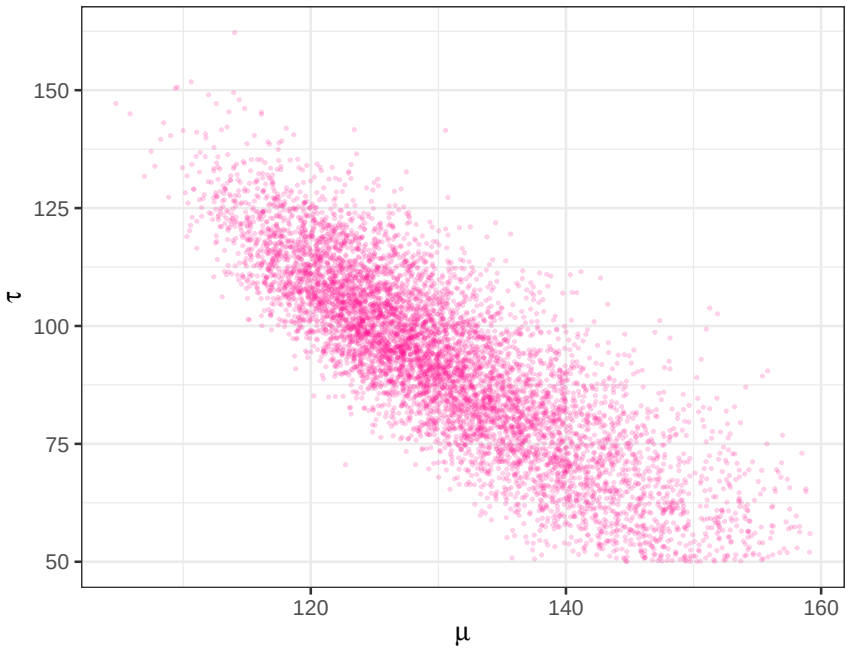


Figure 5: Joint posterior draws of the mean start date and duration of moult (every 20th draw shown).

Question d

```
Running "C:/PROGRA~1/R/R-4.5.3/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.3.0'
gcc  -I"C:/PROGRA~1/R/R-4.5.3/include" -DNDEBUG   -I"C:/Users/isabe/AppData/
    Local/R/win-library/4.5/Rcpp/include/"  -I"C:/Users/isabe/AppData/Local/R/
    win-library/4.5/RcppEigen/include/"  -I"C:/Users/isabe/AppData/Local/R/win-
    library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/isabe/AppData/Local
    /R/win-library/4.5/BH/include" -I"C:/Users/isabe/AppData/Local/R/win-
    library/4.5/StanHeaders/include/src/"  -I"C:/Users/isabe/AppData/Local/R/
    win-library/4.5/StanHeaders/include/"  -I"C:/Users/isabe/AppData/Local/R/
    win-library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/
    Users/isabe/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG
      -DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS
      -DUSE_STANC3 -DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION
    -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/isabe/AppData/Local/R/win-library
    /4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:
    /rtools45/x86_64-w64-mingw32.static.posix/include"      -O2 -Wall -std=
    gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but
     not for C
In file included from C:/Users/isabe/AppData/Local/R/win-library/4.5/RcppEigen
    /include/Eigen/Core:19,
                 from C:/Users/isabe/AppData/Local/R/win-library/4.5/RcppEigen
    /include/Eigen/Dense:1,
                 from C:/Users/isabe/AppData/Local/R/win-library/4.5/
    StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
                 from <command-line>:
```

```
C:/Users/isabe/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file or directory
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-4.5.3/etc/x64/Makeconf:295: foo.o] Error 1
```

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 0.000129 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 1.29 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:     1 / 52000 [  0%]  (Warmup)
Chain 1: Iteration:  2001 / 52000 [  3%]  (Sampling)
Chain 1: Iteration:  7200 / 52000 [ 13%]  (Sampling)
Chain 1: Iteration: 12400 / 52000 [ 23%]  (Sampling)
Chain 1: Iteration: 17600 / 52000 [ 33%]  (Sampling)
Chain 1: Iteration: 22800 / 52000 [ 43%]  (Sampling)
Chain 1: Iteration: 28000 / 52000 [ 53%]  (Sampling)
Chain 1: Iteration: 33200 / 52000 [ 63%]  (Sampling)
Chain 1: Iteration: 38400 / 52000 [ 73%]  (Sampling)
Chain 1: Iteration: 43600 / 52000 [ 83%]  (Sampling)
Chain 1: Iteration: 48800 / 52000 [ 93%]  (Sampling)
Chain 1: Iteration: 52000 / 52000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 1.045 seconds (Warm-up)
Chain 1:                26.118 seconds (Sampling)
Chain 1:                27.163 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2: Rejecting initial value:
Chain 2:   Log probability evaluates to log(0), i.e. negative infinity.
Chain 2:   Stan can't start sampling from this initial value.
Chain 2: Rejecting initial value:
Chain 2:   Log probability evaluates to log(0), i.e. negative infinity.
Chain 2:   Stan can't start sampling from this initial value.
Chain 2: Rejecting initial value:
Chain 2:   Log probability evaluates to log(0), i.e. negative infinity.
Chain 2:   Stan can't start sampling from this initial value.
Chain 2:
Chain 2: Gradient evaluation took 4.6e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.46 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:     1 / 52000 [  0%]  (Warmup)
Chain 2: Iteration:  2001 / 52000 [  3%]  (Sampling)
Chain 2: Iteration:  7200 / 52000 [ 13%]  (Sampling)
```

```
Chain 2: Iteration: 12400 / 52000 [ 23%]  (Sampling)
Chain 2: Iteration: 17600 / 52000 [ 33%]  (Sampling)
Chain 2: Iteration: 22800 / 52000 [ 43%]  (Sampling)
Chain 2: Iteration: 28000 / 52000 [ 53%]  (Sampling)
Chain 2: Iteration: 33200 / 52000 [ 63%]  (Sampling)
Chain 2: Iteration: 38400 / 52000 [ 73%]  (Sampling)
Chain 2: Iteration: 43600 / 52000 [ 83%]  (Sampling)
Chain 2: Iteration: 48800 / 52000 [ 93%]  (Sampling)
Chain 2: Iteration: 52000 / 52000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 1.086 seconds (Warm-up)
Chain 2:                24.302 seconds (Sampling)
Chain 2:                25.388 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 4.8e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take
    0.48 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:     1 / 52000 [  0%]  (Warmup)
Chain 3: Iteration:  2001 / 52000 [  3%]  (Sampling)
Chain 3: Iteration:  7200 / 52000 [ 13%]  (Sampling)
Chain 3: Iteration: 12400 / 52000 [ 23%]  (Sampling)
Chain 3: Iteration: 17600 / 52000 [ 33%]  (Sampling)
Chain 3: Iteration: 22800 / 52000 [ 43%]  (Sampling)
Chain 3: Iteration: 28000 / 52000 [ 53%]  (Sampling)
Chain 3: Iteration: 33200 / 52000 [ 63%]  (Sampling)
Chain 3: Iteration: 38400 / 52000 [ 73%]  (Sampling)
Chain 3: Iteration: 43600 / 52000 [ 83%]  (Sampling)
Chain 3: Iteration: 48800 / 52000 [ 93%]  (Sampling)
Chain 3: Iteration: 52000 / 52000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 1.083 seconds (Warm-up)
Chain 3:                33.351 seconds (Sampling)
Chain 3:                34.434 seconds (Total)
Chain 3:
```

Table 5: Posterior summaries for the Type 3 model obtained using Stan (NUTS), based on 3 chains of 50 000 post-warmup draws.

	Mean	SD	2.5%	Median	97.5%	ESS	$\hat{R}$
μ	130.75	9.53	115.05	129.71	151.50	41715	1
σ^2	317.93	85.52	194.22	303.70	524.47	63042	1
τ	92.50	19.15	55.25	93.35	128.13	45646	1
σ	17.68	2.30	13.94	17.43	22.90	65274	1

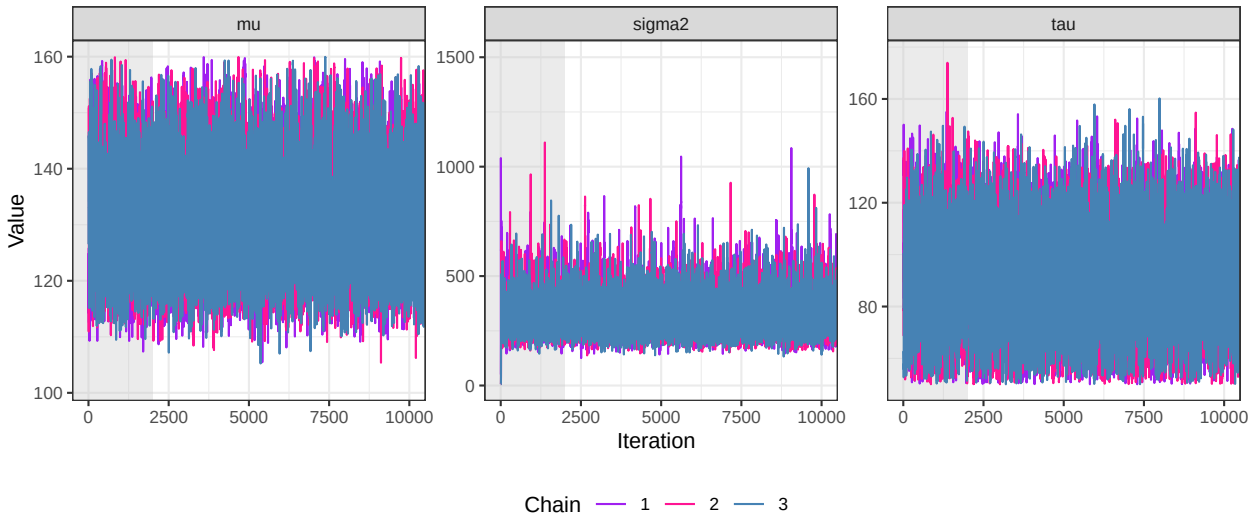


Figure 6: Trace plots of the three Stan chains for the first 4 000 iterations. The shaded region is the warmup period (2 000 iterations), which is discarded.